3tierapplicationdeplyDevSecOps– Project

Step- 1. Create EC2 machine

Clone git repo - https://github.com/VinayKumar1379/3tierapplicationdeplyDevSecOps.git

Go to repo path

Chmod +x *.sh - permission for all .sh file , ./file.sh – use to run

Step- 2. Install jenkin

https://www.jenkins.io/doc/book/installing/linux/

- **4 GB+ of RAM**
- **50 GB+ of drive space**

```
sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
```

Long Term Support release

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install Jenkins
```

Run ./ 1st-Jenkins.sh

```
#!/bin/bash

# Install JRE
sudo apt install openjdk-17-jre-headless -y

# Create keyrings folder
sudo mkdir -p /usr/share/keyrings

# Download Jenkins GPG key
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian/jenkins.io-2023.key

# Add Jenkins repo
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
  https://pkg.jenkins.io/debian binary/" | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null

# Update
sudo apt-get update -y

# Install Jenkins
sudo apt-get install jenkins -y
```

Step- 3. Open all required port

8080 --For Jenkins , 9000 --for SonarQube , 80, 443 , -- 5000 --For Application , 3000 – For DataBase , 22

Now open Jenkins and install all required plugin , create own user id and password

Step- 4. Install Docker
apt install docker.io or
Run Given Script in respo, run ./ 2nd-Docker.sh

```
#!/bin/bash

# Update package manager repositories
sudo apt-get update

# Install necessary dependencies
sudo apt-get install -y ca-certificates curl

# Create directory for Docker GPG key
sudo install -m 0755 -d /etc/apt/keyrings

# Download Docker's GPG key
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc

# Ensure proper permissions for the key
sudo chmod a+r /etc/apt/keyrings/docker.asc

# Add Docker repository to Apt sources
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

# Update package manager repositories
sudo apt-get update

sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-
plugin
```

Step- 5. Add user , integrate all uses like python and Jenkins user access for docker

./3rd-Adduser+python.sh

```
# Add jenkins user to docker group
sudo usermod -aG docker jenkins

# Restart Jenkins to apply changes
sudo systemctl restart jenkins

# Verify Jenkins can access Docker
sudo -u jenkins docker ps

# Update & Install Python pip
sudo apt-get update
sudo apt-get install -y python3-venv python3-pip

# Install docker-compose on your Jenkins server:
```

```
sudo apt-get update
sudo apt-get install -y docker-compose-plugin
```

Step- 6. Install TRIVY

https://trivy.dev/docs/latest/getting-started/installation/

Run ./ 4th-Trivy.sh

```
#!/bin/bash
sudo apt-get install wget apt-transport-https gnupg
wget -qO - https://aquasecurity.github.io/trivy-repo/deb/public.key | gpg --dearmor | sudo tee
/usr/share/keyrings/trivy.gpg > /dev/null
echo "deb [signed-by=/usr/share/keyrings/trivy.gpg] https://aquasecurity.github.io/trivy-repo/deb
generic main" | sudo tee -a /etc/apt/sources.list.d/trivy.list
sudo apt-get update
sudo apt-get install trivy
```

Step- 7. Connect to Docker_Hub

Docker login -u user_name

Password ********

Step -8 Run SonarQube Container

docker run -d ---name sonar -p 9000:9000 sonarqube:lts-community

bydefault, login_id – admin , password – admin

Step -9 Jenkins Configuration

Login Jenkins → settings → Tools →

Once you install the plugin then the option will get created to configure them in the Jenkins

Login Jenkins → Manage jenkins → plugins → Available Plugins → need to install below plugins

1. **SonarQuibe Scanner 2. Docker 3. Docker commons 4. Docker pipeline 5. Docker api 6.docker build step 6. pipeline stage view**
   After install then restart Jenkins server

Step -10 Docker & SonarQube Configuration in Jenkins

SonarQube scanner installation

Jenkins→ Manage →configure tools →

→ Add SonarQube Scanner → Name: Sonar_Scanner, Check install automatically
Select the latest version 7.1.0.4889

Docker installation

→ Add Docker → Name :Docker , check install automatically → click oin add installer, select download from docker.com, version:latest

Step -10 Docker_Hub Credentials Configuration

Manage →credentials →

Security →ceredentials →, click on global , → add credentials →

→ username with password → next → put docker_Hub Global, username: password, ID : (d should be small latter) docker, description :docker

Step -11 System configuration in SonarQube For Jenkins

Login sonarQube – sonarQube work free of cost , but we are going to create token to use SonarQube e logics

Token Creation

Go to SonarQube console --> Administrator → security → users → you can see tokens → Click on 3 dash icon →

A now dialogue → Name : token (T should be in small)→ Expires in :30, 90 days , 1 years, no expiration → generate → you can see token in green colour . copy it Token: squ_amokpokfvlkbjjuhiookjxpfjhh

Now Add The Tokens In Jenkins

Jenkins → settings →Credentials → Global → Add Credentials →Choose : Secret text → Next →
(Secret – past token , ID – sonar_token , Description -sonar-token) → create

SonarQube webhooks creations

SonarQube → Administration → configuration, dropdown →
Webhooks → click on create → name : Jenkins, url – own jenkins login url –
url ex- http://43.204.235.10:8080/sonarqube-webhook/,

Note – Webhooks – jenkins to sonarqube , one env to another env, url and API request via webhooks, url and secret go to inside sonarquebe and give the result

SonarQube Server Configuration in Jenkins

Manage Jenkins → System configuration → system → scroll down to sonarQube server → Add SonarQube →Name: sonar, server url :- url ex- http://43.204.235.20:9000 , Server Athentication token : sonar token, →saved

Step -12 Job Creation in Jenkins pipeline and run

Manual Docker image testing using Trivy

Installation trivy - https://trivy.dev/docs/latest/getting-started/installation/

Trivy image image_name:latest